A quartic counterexample to diagonal concavity and a real-rootedness criterion

Research proof and formalization packet
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Abstract

We give one explicit quartic construction refuting two conjectures of Lachaume. The polynomial $P(x) = x^4 + 8x^3 + 6x^2 + 8x + 1$ has positive coefficients and satisfies $4PP'' - 3(P')^2 \le 0$ on the entire real line, but has a nonreal root. The corresponding symmetric polynomial $f = 1 + 2\sigma_1 + \sigma_2 + 2\sigma_3 + \sigma_4$ in four variables has a concave fourth root on the positive diagonal, while an exact midpoint inequality shows that $f^{1/4}$ is not concave on the positive orthant. All inputs and all five coefficients are strictly positive. The complete statements, including the genuine elementary symmetric functions and the universal diagonal-concavity assertion, are certified in Lean.

1 The source questions and the construction

For $x = (x_1, \dots, x_n)$ write

$$\sigma_k(x) = \sum_{1 \le i_1 < \cdots < i_k \le n} x_{i_1} \cdots x_{i_k}, \qquad \sigma_0(x) = 1.$$

Let $\Gamma_n = (0, \infty)^n$. Lachaume [1, Conjecture 1, p. 3] asks whether, for $a_0, \dots, a_p \ge 0$, concavity of $\left(\sum_{k=0}^{p} a_k \sigma_k\right)^{1/p}$ on Γ_n is equivalent to concavity of its restriction to the positive diagonal. In the same paper, Conjecture 3 (p. 5) proposes that a polynomial P of degree n with nonnegative coefficients is real-rooted whenever

$$PP'' + \left(\frac{1}{n} - 1\right)(P')^2 \le 0 \qquad \text{on all of } \mathbb{R}. \tag{1}$$

The positive-axis restriction of this condition is insufficient for that conjecture; we verify the displayed condition on the full real line.

Theorem 1. *Set $n = p = 4$, $(a_0, \dots, a_4) = (1, 2, 1, 2, 1)$, and*

$$f = 1 + 2\sigma_1 + \sigma_2 + 2\sigma_3 + \sigma_4, \qquad P(t) = f(t, t, t, t) = t^4 + 8t^3 + 6t^2 + 8t + 1.$$

Then:

1. *P satisfies (1) on $\mathbb{R}$ and has a nonreal root.*
2. *$P^{1/4}$ is concave on $(0, \infty)$.*
3. *$f^{1/4}$ is positive and twice continuously differentiable on Γ_4, but is not concave there.*

Thus this single construction refutes both cited conjectures.

2 The global differential condition and a nonreal root

Differentiating gives

$$P' = 4t^3 + 24t^2 + 12t + 8, \qquad P'' = 12t^2 + 48t + 12.$$

Expansion yields the exact identity

$$4PP'' - 3(P')^2 = -144(t^2 - 1)^2. \tag{2}$$

This proves (1) for every real t, since $n = 4$. All coefficients of P are positive, so $P(t) > 0$ for $t > 0$. For the genuine positive fourth root $h(t) = P(t)^{1/4}$,

$$h''(t) = \frac{4P(t)P''(t) - 3P'(t)^2}{16P(t)^{7/4}} \le 0 \qquad (t > 0).$$

Consequently h is concave on the full positive axis.

To exhibit a nonreal root, put $s = \sqrt{3}$. The factorization

$$P(t) = \big(t^2 + (4-2s)t + 1\big)\big(t^2 + (4+2s)t + 1\big) \tag{3}$$

follows from $s^2 = 3$. Since $3/2 < s < 2$, the number

$$z = s - 2 + i\sqrt{4s-6}$$

has strictly positive imaginary part. Substitution into the first factor in (3) gives zero: writing $u = s-2$, $v = \sqrt{4s-6}$, the imaginary part is $v(2u+4-2s) = 0$, and the real part is $u^2 - v^2 + (4-2s)u + 1 = 0$. Therefore $P(z) = 0$, proving the first two assertions of Theorem 1.

3 A strictly positive midpoint witness

Fix the first coordinate at 2. Direct expansion of the elementary symmetric functions gives

$$g(t) := f(2,t,t,t) = 5 + 12t + 15t^2 + 4t^3. \tag{4}$$

Consider the strictly positive points

$$A = (2, 1/20, 1/20, 1/20), \quad B = (2, 3/20, 3/20, 3/20), \quad C = \frac{A+B}{2} = (2, 1/10, 1/10, 1/10).$$

Equation (4) gives

$$f(A) = \frac{2819}{500}, \qquad f(C) = \frac{3177}{500}, \qquad f(B) = \frac{7151}{1000}.$$

The following are exact rational inequalities:

$$\left(\frac{15409}{10000}\right)^4 < \frac{2819}{500}, \qquad \frac{3177}{500} < \left(\frac{15877}{10000}\right)^4, \qquad \left(\frac{16352}{10000}\right)^4 < \frac{7151}{1000}.$$

Monotonicity of the fourth power on nonnegative numbers now proves

$$f(C)^{1/4} < \frac{15877}{10000} < \frac{1}{2}\left(\frac{15409}{10000} + \frac{16352}{10000}\right) < \frac{f(A)^{1/4} + f(B)^{1/4}}{2}.$$

The middle rational gap is $7/20000$. This strictly reverses the midpoint inequality required by concavity. Moreover $f > 0$ on Γ_4 because all its coefficients and coordinates are positive; hence its fourth root is smooth there. This completes the proof of Theorem 1.

4 Formal certificate and scope

The accompanying three Lean modules contain 61 declarations, replayed from fresh source copies with Lean 4.34.1 and mathlib revision `d13f23b7`. The verifier treats warnings as errors and checks the axioms of every declaration; only `propext`, `Classical.choice`, and `Quot.sound` occur. The two source endpoints include the actual polynomial and its derivatives, a nonreal root of its complex extension, strict positivity and C^2 regularity, universal diagonal concavity, and failure of concavity on the positive orthant. The elementary symmetric functions are literal subset sums, identified with Mathlib's `MvPolynomial.esymm`. No numerical approximation is used in the certificate.

The distinct real-rooted-diagonal sufficient condition is not refuted. Lachaume's later work [2, Theorem 2.7(d)] establishes that condition using Walsh's theorem. Carptopus [3] identifies the derivative convolution in the original Conjectures 4 and 5 with a rescaled Walsh convolution and explicitly excludes Conjectures 1 and 3 from its scope. We claim neither a new Walsh theorem nor a resolution of those neighboring questions. The two failures here form one common construction. The independent source and prior-art audit is pending; this draft does not assert worldwide priority or a complete literature census.

References

[1] X. Lachaume, *On the concavity of a sum of elementary symmetric polynomials*, arXiv:1712.10327v1 (2017), https://arxiv.org/abs/1712.10327.

[2] X. Lachaume, *The constraint equations of Lovelock gravity theories: a new σ_k-Yamabe problem*, J. Math. Phys. **59**, 072501 (2018), arXiv:1712.04528v2, doi:10.1063/1.5023758.

[3] Carptopus, *A Walsh-convolution proof of two conjectures of Lachaume*, v0.1-beta (4 September 2026), doi:10.5281/zenodo.22299983.